05-Retrieval-Architecture.md

Healthcare Clinical Intelligence Platform

Advanced Retrieval Architecture

Version: 1.0

Status: Approved Design

Mission

Build a healthcare-grade retrieval system that is:

- Self-Correcting
- Cost Efficient
- Highly Accurate
- Fault Tolerant
- Fully Observable
- Governance Aware
- Multi-Tenant
- Production Ready

The retrieval system should maximize evidence quality while minimizing cost and hallucinations.

Retrieval Philosophy

Bad RAG:

```
Retrieve
↓
Answer
```

Our Architecture:

```
Understand Query
↓
Plan Retrieval
```

```
↓
Retrieve
↓
Validate Evidence
↓
Confidence Check
↓
Corrective Retrieval
↓
Context Optimization
↓
Response Generation
```

Retrieval Stack

Layer 1

Query Understanding

Layer 2

Retrieval Planning

Layer 3

Retrieval Execution

Layer 4

Evidence Validation

Layer 5

Corrective Retrieval

Layer 6

Context Optimization

Layer 7

Response Consumption

Query Understanding Engine

Responsibilities:

- Intent Classification
- Query Complexity Detection
- Source Selection
- Risk Classification

Query Types

Clinical

Drug Interaction

Research

Policy

Insurance

Patient Education

Administrative

Complexity Levels

Level 1

Simple Lookup

Level 2

Multi-source Retrieval

Level 3

Reasoning Heavy

Level 4

Research Grade

Query Planner

Output Example

```
{
  "query_type":"clinical",
  "complexity":"high",
  "sources":[
    "hospital_kb",
    "pubmed",
    "fda"
  ],
  "retrieval_mode":"hybrid"
}
```

Retrieval Sources

Priority Order

```
Organization KB
↓
Department KB
↓
Personal KB
↓
External Knowledge Base
```

```
↓
Live Search
```

Retrieval Types

Dense Retrieval

Qdrant

Used For:

- Semantic Search
- Similar Concepts
- Research Questions

Sparse Retrieval

Elasticsearch

Used For:

- Exact Matches
- Drug Names
- Codes
- Policies

Graph Retrieval

Neo4j

Used For:

- Disease Relationships
- Drug Interactions
- Treatment Paths

Metadata Retrieval

Filters:

- Organization
- Department
- Role
- Version
- Approval Status

Hybrid Retrieval

Default Strategy

```
Vector Search
+
BM25
+
Metadata Filters
```

Graph Retrieval added only when necessary.

Parent Child Retrieval

Structure

```
Guideline
     |
Recommendation
     |
Clinical Detail
```

Retrieve child chunks.

Return parent context.

Improves reasoning quality.

Query Rewriting Engine

Example

User:

Can diabetic patients receive contrast CT?

Rewrites:

Metformin contrast CT

Diabetes imaging contrast safety

FDA contrast guideline

PubMed contrast studies

Multi Query Retrieval

Generate:

Original Query

Clinical Variant

Research Variant

Keyword Variant

Ontology Variant

Retrieve from all.

Merge results.

GraphRAG Trigger Engine

GraphRAG only activated for:

Drug Questions

Disease Questions

Treatment Questions

Clinical Pathways

Research Questions

GraphRAG Example

```
Metformin
      |
Treats
      |
Type 2 Diabetes
      |
Associated With
      |
Kidney Disease
```

Used to expand evidence coverage.

Retrieval Pipeline

```
Query
 ↓
Planner
 ↓
Query Rewriter
 ↓
Multi Query Generator
 ↓
Hybrid Retrieval
 ↓
Merge Results
 ↓
Reranking
 ↓
Validation
 ↓
Confidence Engine
```

```
  ↓
Context Builder
```

Initial Retrieval

Retrieve:

50-100 chunks

Across all sources.

Never send directly to LLM.

Reranking Layer

Model:

BGE Reranker

Future:

Cohere Rerank

Flow

```
100 Chunks
      ↓
Reranker
      ↓
Top 15 Chunks
```

Evidence Validation Engine

Checks:

- Source Quality

• Citation Availability
• Evidence Coverage
• Contradictions
• Freshness

Source Quality Scores

Hospital SOP

95

FDA

95

WHO

95

CDC

90

PubMed

90

Personal Notes

70

Source Diversity Engine

Bad

10 chunks from one PDF

Good

Hospital SOP

PubMed

FDA

WHO

Confidence Engine

Inputs:

Retrieval Score

Rerank Score

Source Diversity

Freshness

Evidence Coverage

Output

```
{
  "confidence":0.91,
  "status":"sufficient"
}
```

Self Correcting RAG

Activated when:

Confidence < Threshold

Missing Evidence

Contradictory Evidence

Corrective Retrieval Loop

```
Retrieve
  ↓
Validate
  ↓
Confidence Low?
  ↓
Yes
  ↓
Rewrite Query
  ↓
Expand Query
  ↓
Retrieve Again
  ↓
Validate Again
```

Maximum:

3 Retrieval Attempts

Escalation Retrieval

If confidence still low:

```
PubMed
  ↓
FDA
  ↓
WHO
  ↓
Live Search
```

Healthcare Safety Rule

If confidence remains low:

DO NOT generate definitive medical advice.

Return:

```
Evidence insufficient.

Consult specialist review.
```

Context Optimization Layer

Before LLM

```
15 Chunks
 ↓
Compression
 ↓
8 Optimized Chunks
```

Context Compression

Remove:

Duplicate Evidence

Redundant Text

Low Value Chunks

Cost Optimization Engine

Cheap Path

BM25

Vector Search

Medium Path

Hybrid Search

Reranking

Expensive Path

GraphRAG

External Search

Corrective Retrieval

Only used when needed.

Caching Strategy

Query Cache

Redis

Retrieval Cache

Top Retrieved Chunks

Semantic Cache

Embedding Similarity

External Source Cache

PubMed Results

FDA Results

WHO Results

Fault Tolerance

Qdrant Failure

Fallback:

BM25

Neo4j Failure

Fallback:

Vector Search

External API Failure

Fallback:

Cached Data

Retrieval Guardrails

Must Enforce:

RBAC

Department Access

Tenant Isolation

Document Approval

Document Version

Retrieval Metrics

Track:

Context Precision

Context Recall

Source Diversity

Retrieval Confidence

Retrieval Latency

Retrieval Cost

Retry Rate

Failure Rate

Observability Events

Emit:

Query Received

Query Rewritten

Retrieval Started

Retrieval Completed

Reranking Completed

Validation Completed

Confidence Calculated

Retry Triggered

Context Generated

LangGraph Nodes

Query Planner

Query Rewriter

Multi Query Generator

Retriever

Graph Retriever

Reranker

Evidence Validator

Confidence Engine

Retry Controller

Context Builder

Success Targets

Context Precision > 90%

Context Recall > 85%

Source Diversity > 3 Sources

Retrieval Latency < 1.5s

Confidence Accuracy > 90%

Retry Rate < 10%

Failure Rate < 1%

Final Principle

The retrieval layer is the brain of the platform.

The LLM should never compensate for poor retrieval.

Retrieval quality, evidence quality, and governance determine the trustworthiness of the entire healthcare AI system.